

Semester Project - Team 2

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CHE 494. Biopharmaceutical Process Development and
Manufacturing

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Working for a Pharmaceutical Company

- Work impact
- Interdisciplinary team work
- Access to cutting edge technology



Drug Product Process Selection

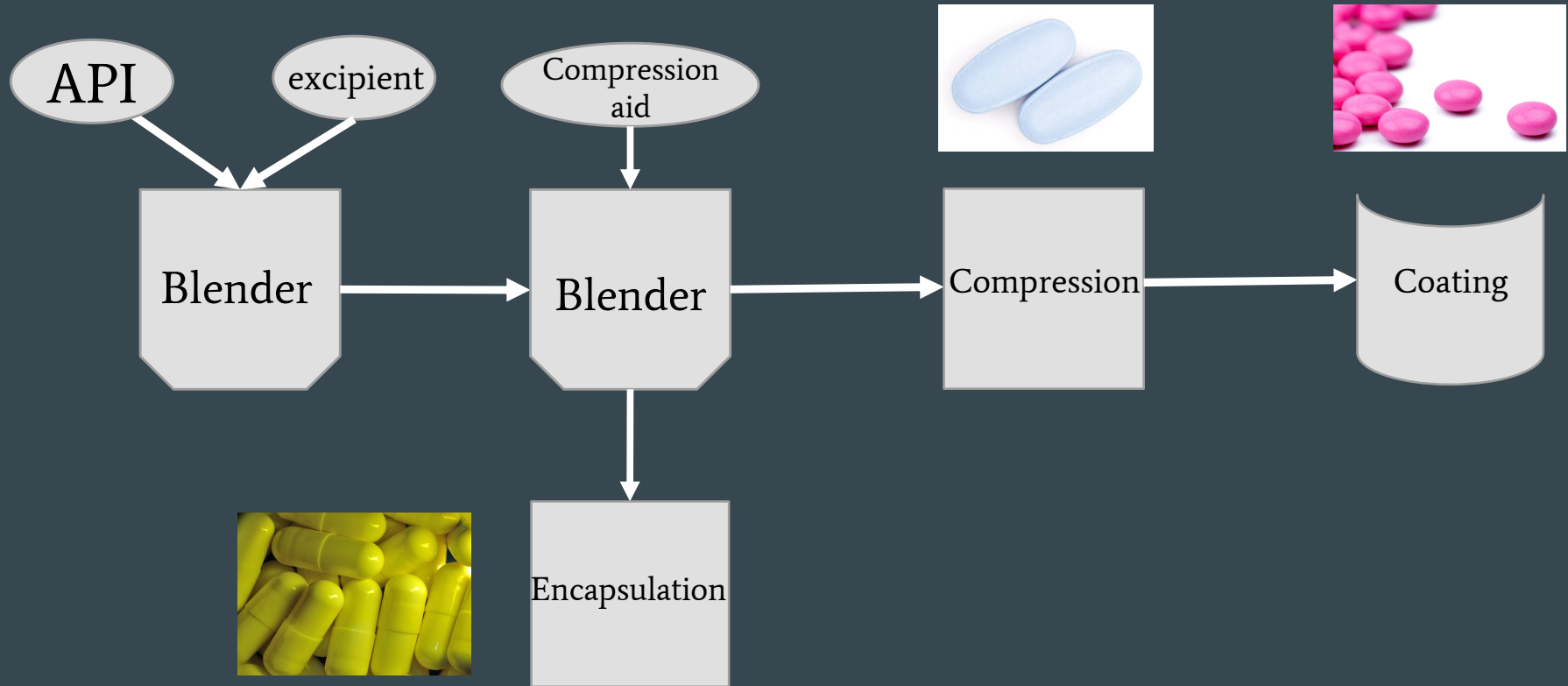
We are given a soluble small molecule substance API with the following characteristics:

1. Tight particle size distribution
2. Great flowability
3. High bulk density

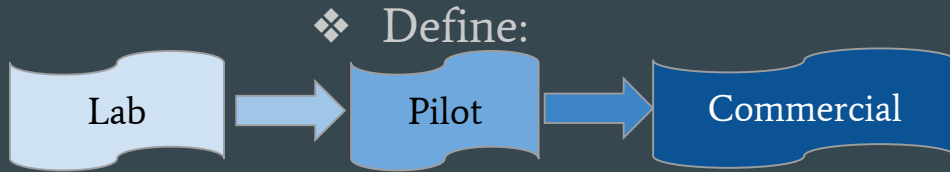


What drug product process would we recommend
for solid dosage form?

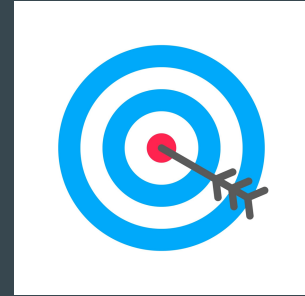
Selected Process: Direct Compression



Reliable Scale-Up Process w/ context



❖ Goal:



❖ Strategies:

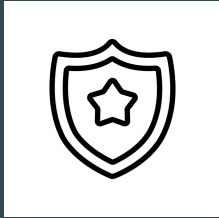
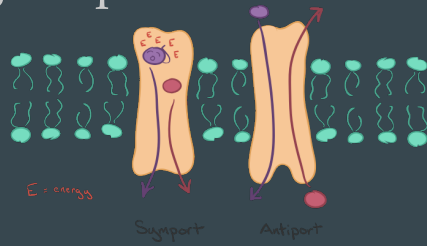


❖ Reproducibility:

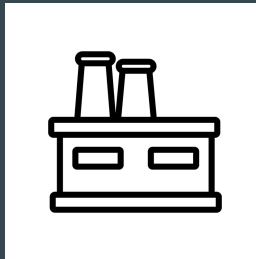
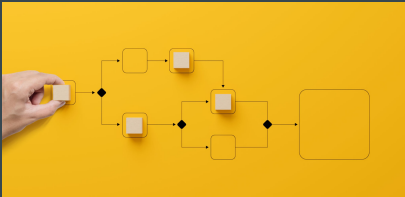


Key considerations

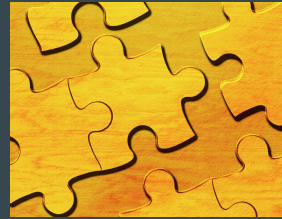
❖ Drug Properties:



❖ Manufacturing:



❖ Formulation:



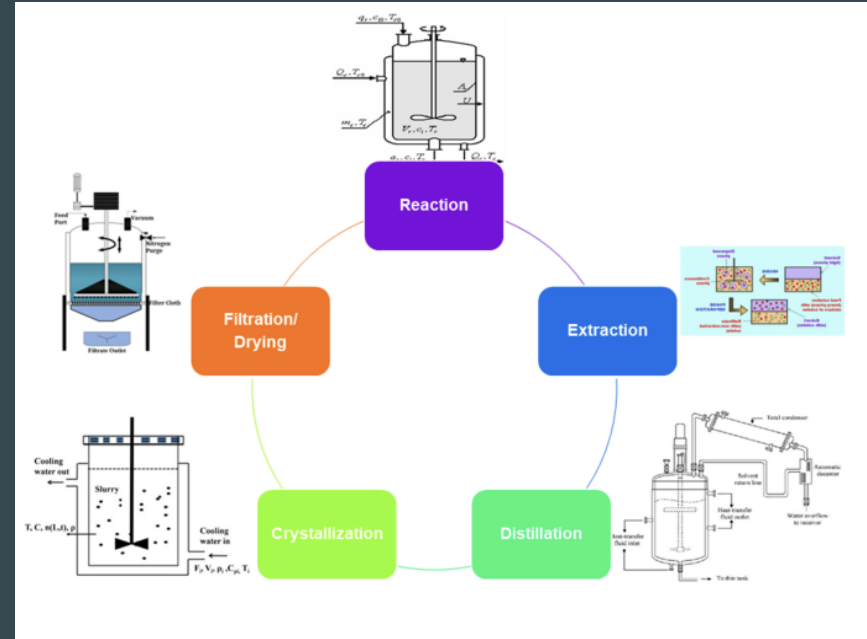
❖ Patient:



Most prevalent unit process in small molecule drug substance process

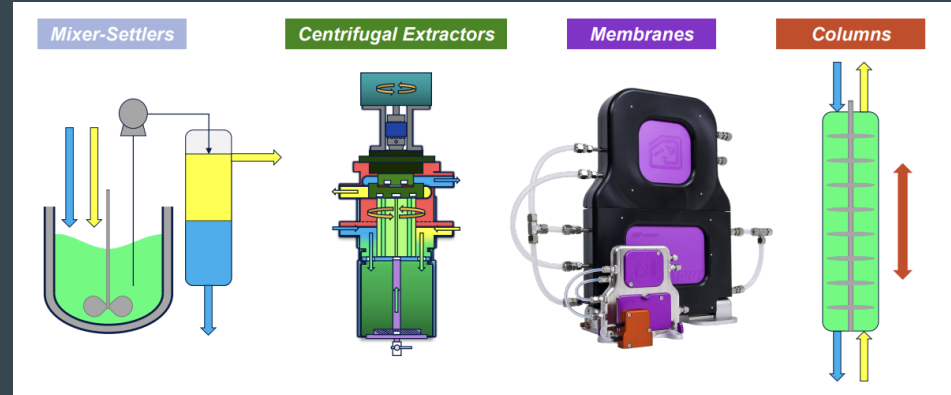
Crystallization is the most prevalent process

- Primary method of purification
 - Selectively incorporates compound into a solid lattice while excluding impurities
- Defines solid-state properties/form
 - Effects particle size distribution
- Scales efficiently
 - Easy to scale from lab to plant



Key considerations for the design of optimal liquid-liquid extraction process

- Distribution Coefficient
 - High K_D → Fewer stages and less solvent needed
- Selectivity
 - High selectivity ensures clean separation
- Type of extractor
 - Effects mass transfer, scalability, and footprint



Reaction Analysis Using a Graph

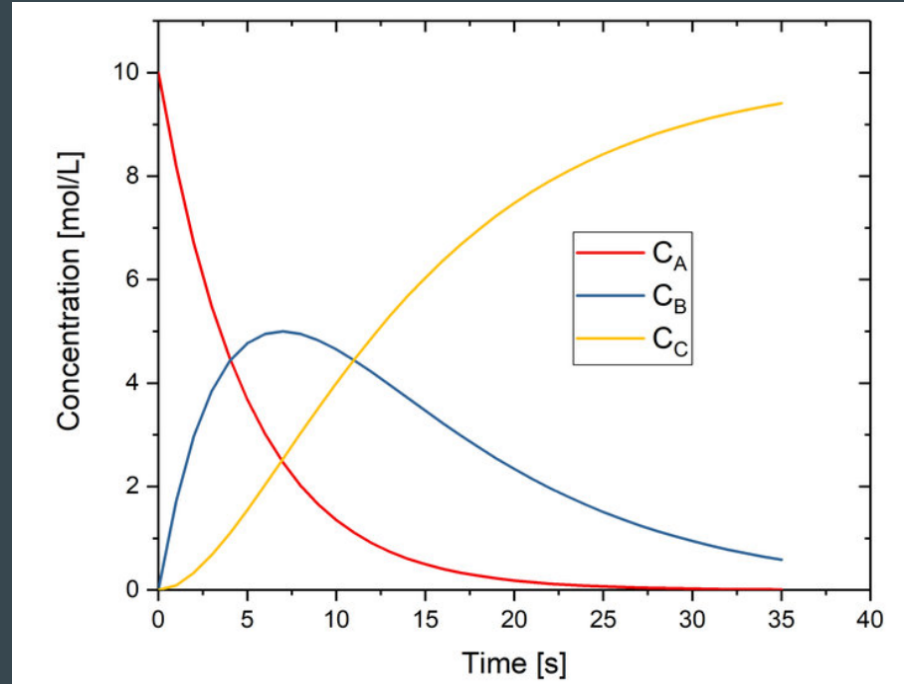
Given this graph, we are asked to identify the reaction mechanism, recommend a reactor for this reaction, and provide recommendations for a reliable reaction scale-up.

Graph Key:

C_A = Reactant

C_B = Product

C_C = Undesired Product



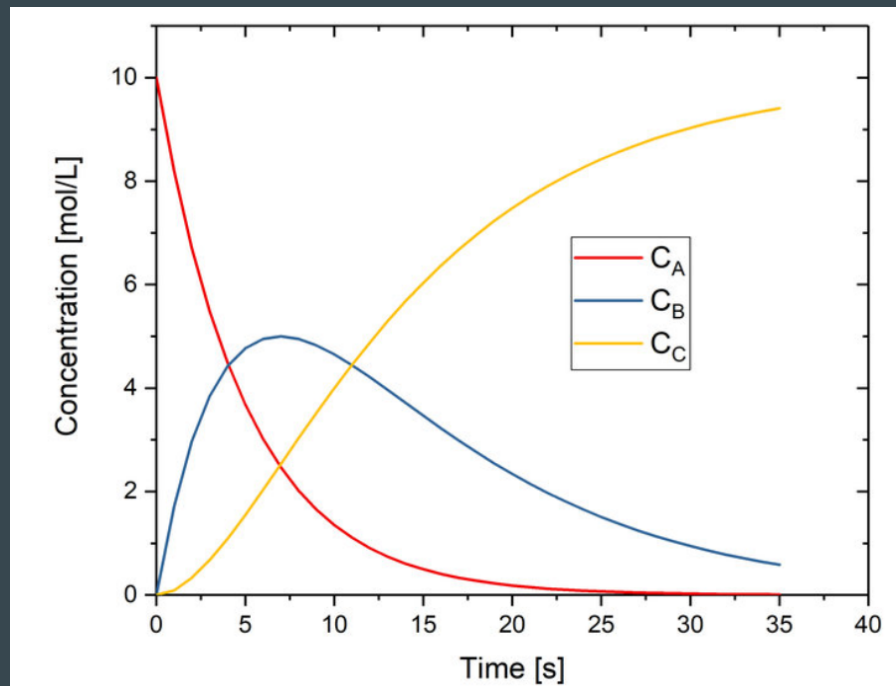
Reaction Analysis

Reaction Type: $A \rightarrow B \rightarrow C$
(Consecutive reaction)

Recommended type of reactor: Batch
Reactor

Scale-up focus:

- Precise time control
- Temperature management

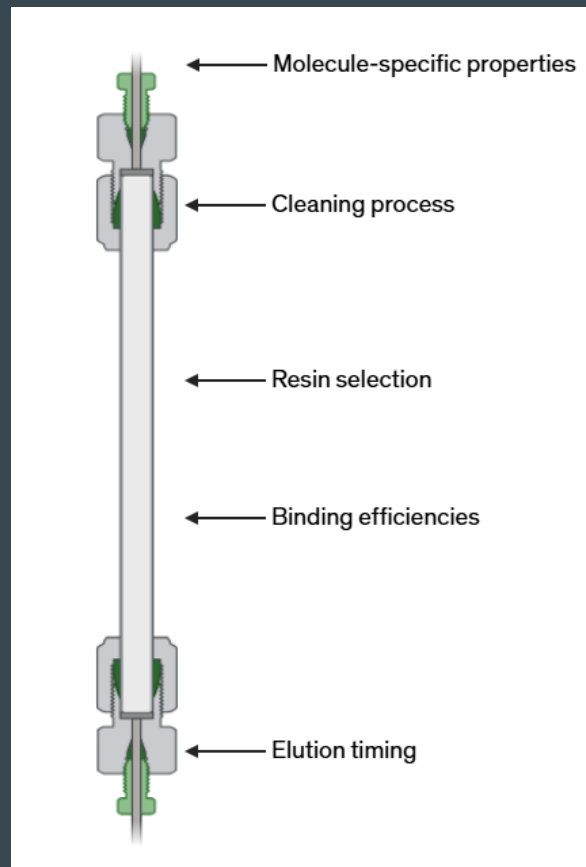
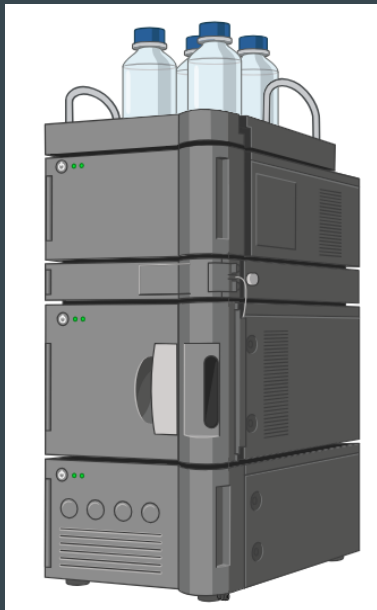


Liquid chromatography: the most difficult purification technique

Many considerations are necessary to conduct liquid chromatography.

Scale-up is difficult as size can affect:

- Residence times
- Column packing
- Flow distribution
- Equipment pressure
- Cleaning process
- Meeting regulatory standards



References

- Lecture Slides